

## Classification Model Summary

Three classification algorithms were developed to classify houses into two price categories:

- Low Price House (0)
- High Price House (1)

The models evaluated were:

1. Logistic Regression
2. Decision Tree Classifier
3. Random Forest Classifier

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

The models were compared based on their performance, with particular attention given to the F1 Score because it provides a balance between precision and recall.

The feature importance analysis from the Random Forest model also provided insights into the house characteristics that contributed most to distinguishing between low- and high-priced houses.

### CONCLUSION

The classification task demonstrated that machine learning can effectively classify houses into price categories using property-related features. Among the three models evaluated, Random Forest performed best, achieving 77.46% accuracy and a 77.14% F1-score. It was therefore selected as the preferred classification model.

Recommendations: Future improvements should focus on hyperparameter tuning, feature engineering, obtaining additional relevant housing data, and testing other classification algorithms. The resulting model can be used as a decision-support tool for preliminary house price-category assessment rather than as a substitute for professional property valuation.